FA4_APM1220_Quijano

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2026-09-17

```
library(ICSNP)
```

```
## Loading required package: mvtnorm
```

```
## Warning: package 'mvtnorm' was built under R version 4.4.3
```

```
## Loading required package: ICS
```

```
## Warning: package 'ICS' was built under R version 4.4.3
```

```
library(poolr)
```

```
## Warning: package 'poolr' was built under R version 4.4.3
```

```
library(car)
```

```
## Warning: package 'car' was built under R version 4.4.3
```

```
## Loading required package: carData
```

```
## Warning: package 'carData' was built under R version 4.4.3
```

1. Hotelling's T^2 Test for a Mean Vector

(a) Evaluate T^2 for testing the null hypothesis $H_0 : \mu^T = [7, 11]$.

```
x <- matrix(c(2, 12, 8, 9, 6, 9, 8, 10), nrow=4, ncol=2, byrow=TRUE)
x
```

```
##      [,1] [,2]
## [1,]    2   12
## [2,]    8    9
## [3,]    6    9
## [4,]    8   10
```

```r
n <- nrow(x)
p <- ncol(x)

t2 <- HotellingsT2(x, mu = c(7, 11))
t2$statistic
```

```
##      T.2
## 4.545455
```

(b) Specify the exact distribution of T^2 under the null hypothesis for this situation.

The exact distribution is:

$\frac{n-p}{p(n-1)}T^2 \sim F_{p,n-p}$

Given n (rows) = 4, p (columns) = 2:

$\frac{4-2}{2(4-1)}T^2 = \frac{T^2}{3}$

Thus, the exact distribution of T^2 under the null hypothesis is:

$T^2 \sim 3F_{2,2}$

(c) Using parts (a) and (b), test H_0 at the significance level $\alpha = 0.05$. What statistical conclusion do you reach?

```r
Fstat <- ((n - p) / (p * (n - 1))) * t2$statistic
Fstat
```

```
##      T.2
## 1.515152
```

```r
# Critical value
qf(0.95, p, n - p)
```

```
## [1] 19
```

```r
# p-value
pf(Fstat, p, n - p, lower.tail = FALSE)
```

```
##       T.2
## 0.3975904
```

With a significance level of $\alpha = 0.05$, we acquire $F_{0.95,2,2} = 19$. Since the F-statistic $= 1.515152 < 19$ we fail to reject H_0 at $\alpha = 0.05$.

2. Simultaneous Confidence Intervals and Regions for Grizzly Bear Body Measurements

```r
n <- 61
p <- 6 # variables in order are: Weight, Body Length, Neck, Girth, Head Length, Head Width

xbar <- c(95.52, 164.38, 55.69, 93.39, 17.98, 31.13)
```

```r
S <- matrix(c(
  3266.46, 1343.97, 731.54, 1175.50, 162.68, 238.37,
  1343.97, 721.91, 324.25, 537.35, 80.17, 117.73,
  731.54, 324.25, 179.28, 281.17, 39.15, 56.80,
  1175.50, 537.35, 281.17, 474.98, 63.73, 94.85,
  162.68, 80.17, 39.15, 63.73, 9.95, 13.88,
  238.37, 117.73, 56.80, 94.85, 13.88, 21.26
), nrow = 6, byrow = TRUE)
```

(a) Obtain the large-sample 95% simultaneous confidence intervals for the six population mean body measurements.

```r
critical <- qchisq(0.95, df=6)
critical
```

```
## [1] 12.59159
```

```r
margin <- sqrt((critical * diag(S))/n)
margin
```

```
## [1] 25.966535 12.207222  6.083328  9.901773  1.433134  2.094869
```

```r
lwr_bound <- xbar - margin
upr_bound <- xbar + margin

cbind(lwr_bound, upr_bound)
```

```
##      lwr_bound upr_bound
## [1,]  69.55347 121.48653
## [2,] 152.17278 176.58722
## [3,]  49.60667  61.77333
## [4,]  83.48823 103.29177
## [5,]  16.54687  19.41313
## [6,]  29.03513  33.22487
```

(b) Obtain the large-sample 95% simultaneous confidence ellipse for mean weight and mean girth.

```r
# Mean Weight = 95.52, Mean Girth = 93.39
x_b <- c(xbar[1], xbar[4])

# Covariance Matrix of weight and girth
s_weight_girth <- matrix(c(
  3266.46, 1175.50,
  1175.50, 474.98),
  nrow=2, ncol=2, byrow=TRUE)
```

The simultaneous confidence ellipse is given by

$$(\mu - \bar{x})' S_{WG}^{-1} (\mu - \bar{x}) \leq \frac{\chi^2_{p,1-\alpha}}{n}$$

```r
# chi squared
critical <- qchisq(0.95, df=2)

rhs <- critical/61
rhs
```

```
## [1] 0.09822073
```

or approximately

$$(\mu - \bar{x})' S_{WG}^{-1} (\mu - \bar{x}) \leq 0.09822073$$

Explicitly, this is

$$\begin{bmatrix} \mu_W - 95.52 \\ \mu_G - 93.39 \end{bmatrix}^T \begin{bmatrix} 3266.46 & 1175.50 \\ 1175.50 & 474.98 \end{bmatrix}^{-1} \begin{bmatrix} \mu_W - 95.52 \\ \mu_G - 93.39 \end{bmatrix} \leq 0.09822073$$

(c) Obtain the 95% Bonferroni confidence intervals for the six means in part (a).

```r
z_bonf <- qnorm(1 - 0.05/(2*6))
z_bonf
```

```
## [1] 2.638257
```

```r
margin_bonf <- z_bonf * sqrt(diag(S)/n)

lower_bonf <- xbar - margin_bonf
upper_bonf <- xbar + margin_bonf

bonf_intervals <- cbind(lower_bonf, upper_bonf)
bonf_intervals
```

```
##      lower_bonf upper_bonf
## [1,]   76.21406  114.82594
## [2,]  155.30402  173.45598
## [3,]   51.16709   60.21291
## [4,]   86.02810  100.75190
## [5,]   16.91447   19.04553
## [6,]   29.57248   32.68752
```

(d) Refer to part (b). Construct the 95% Bonferroni confidence rectangle for mean weight and mean girth using m = 6. Compare this rectangle with the simultaneous confidence ellipse in part (b).

With m = 6, we still use $z^* = 2.638257$. Thus, for weight

$$95.52 \pm 2.638257 \sqrt{\frac{3266.46}{61}} 76.21406 < \mu_W < 114.82594$$

and girth

$$93.39 \pm 2.638257 \sqrt{\frac{474.98}{61}} 86.02810 < \mu_G < 100.75190$$

(e) Obtain the 95% Bonferroni confidence interval for (mean head width - mean head length) using m = 6 + 1 = 7 to allow for this statement as well as statements about each individual mean.

```
difference <- xbar[6] - xbar[5]

variance <- S[6,6] + S[5,5] - 2*S[6,5]

SE <- sqrt(variance / n)

m <- 7

z <- qnorm(1 - 0.05/(2*m))

lower <- difference - z*SE
upper <- difference + z*SE

c(lower, upper)
```

```
## [1] 12.51024 13.78976
```

3. Mineral Analysis of Ancient Peruvian Hairs

```
Cr <- c(.48, 40.53, 2.19, .55, .74, .66, .93, .37, .22)

St <- c(12.57, 73.68, 11.13, 20.03, 20.29, .78, 4.64, .43, 1.08)
n = 9

hair <- cbind(Cr, St)

hair
```

```
##          Cr    St
##  [1,]  0.48 12.57
##  [2,] 40.53 73.68
##  [3,]  2.19 11.13
##  [4,]  0.55 20.03
##  [5,]  0.74 20.29
##  [6,]  0.66  0.78
##  [7,]  0.93  4.64
##  [8,]  0.37  0.43
##  [9,]  0.22  1.08
```

(a) Construct and plot a 90% joint confidence ellipse for the population mean vector $\mu^T = [\mu_1, \mu_2]$, assuming these nine hairs represent a random sample from a particular ancient Peruvian culture.

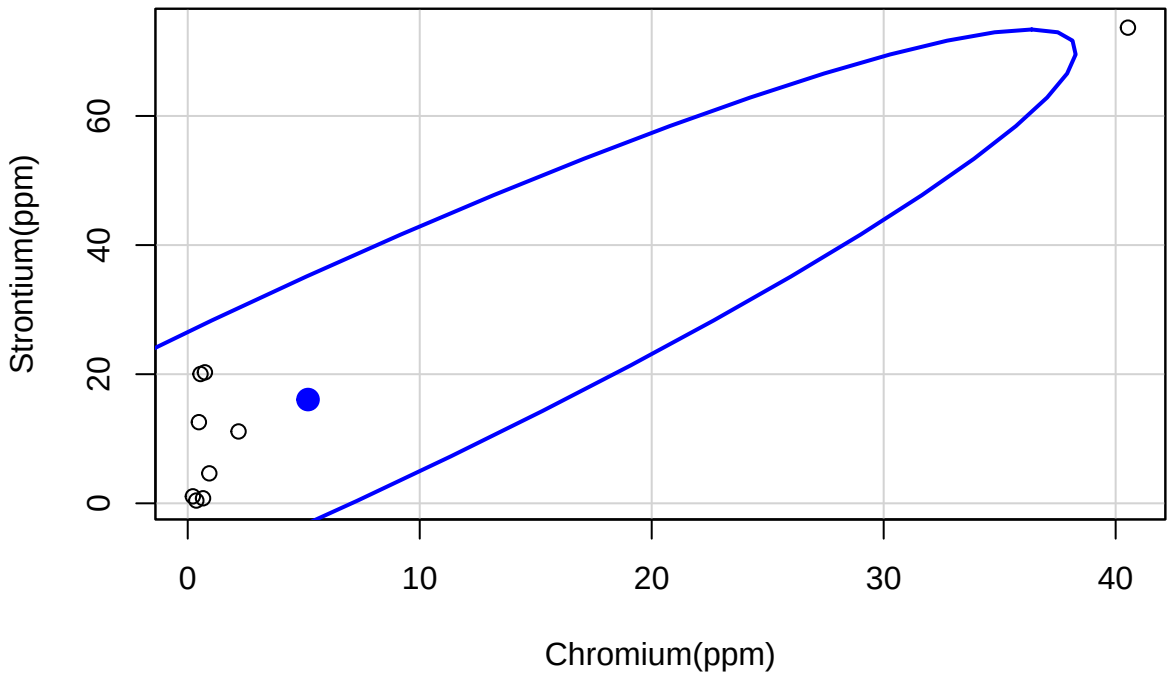
```
x_bar <- colMeans(hair)
x_bar
```

```
##        Cr        St
##  5.185556 16.070000
```

```r
s <- cov(hair)

critical <- qchisq(0.90, df=2)

dataEllipse(
  hair[,1],
  hair[,2],
  levels = 0.90,
  xlab = "Chromium(ppm)",
  ylab = "Strontium(ppm)"
)
```



(b) Obtain the individual simultaneous 90% confidence intervals for μ_1 and μ_2 by projecting the ellipse constructed in part (a) onto each coordinate axis. Does it appear as if this Peruvian culture has a mean strontium level of 10? That is, are any of the points (μ_1, arbitrary, 10) in the confidence region? Is [.30, 10]' a plausible value for μ? Discuss.

```r
margin <- sqrt(
  qchisq(0.90, df = 2) * diag(s) / n
)

lower <- x_bar - margin
upper <- x_bar + margin

cbind(lower, upper)
```

```
##         lower    upper
## Cr -4.3043759 14.67549
## St -0.3645025 32.50450
```

Yes, it is plausible with a 90% confidence interval that a point exists with strontium = 10 as it is within the interval. As for [.30, 10],

```
mu0 <- c(0.30,10)

d <- mu0 - x_bar

t <- n * t(d) %*% solve(s) %*% d

t
```

```
##          [,1]
## [1,] 1.772481
```

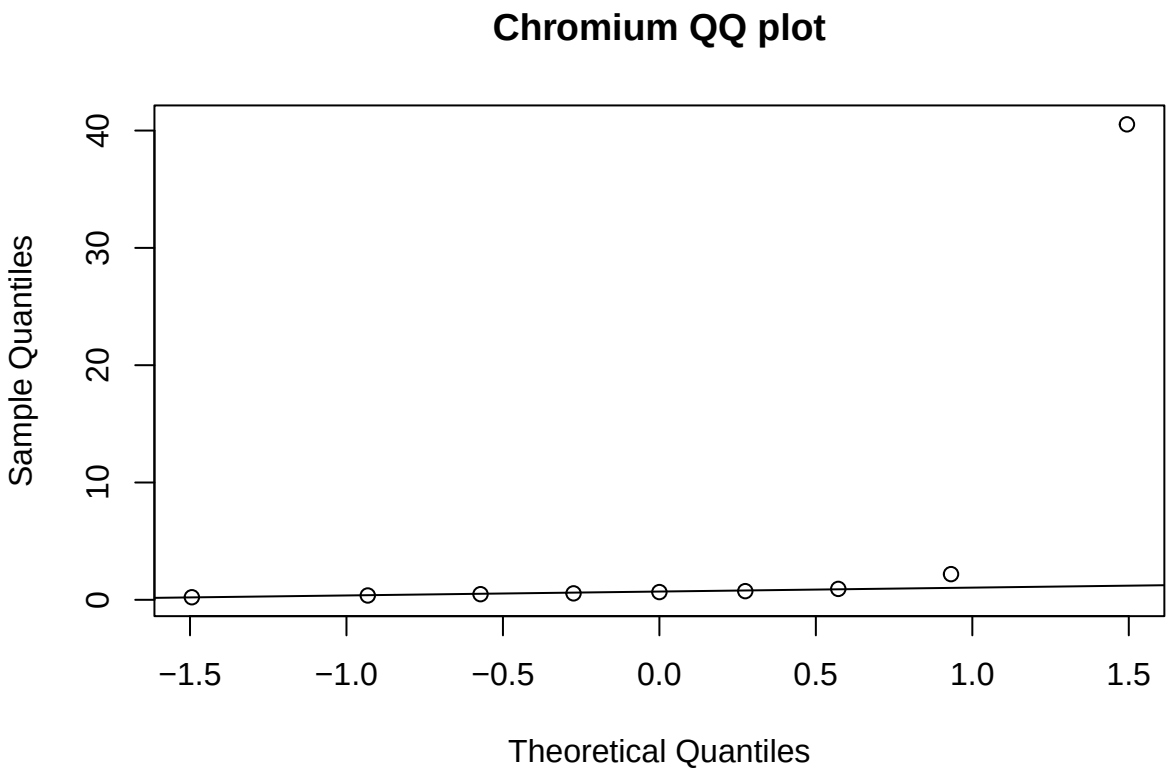
```
critical
```

```
## [1] 4.60517
```

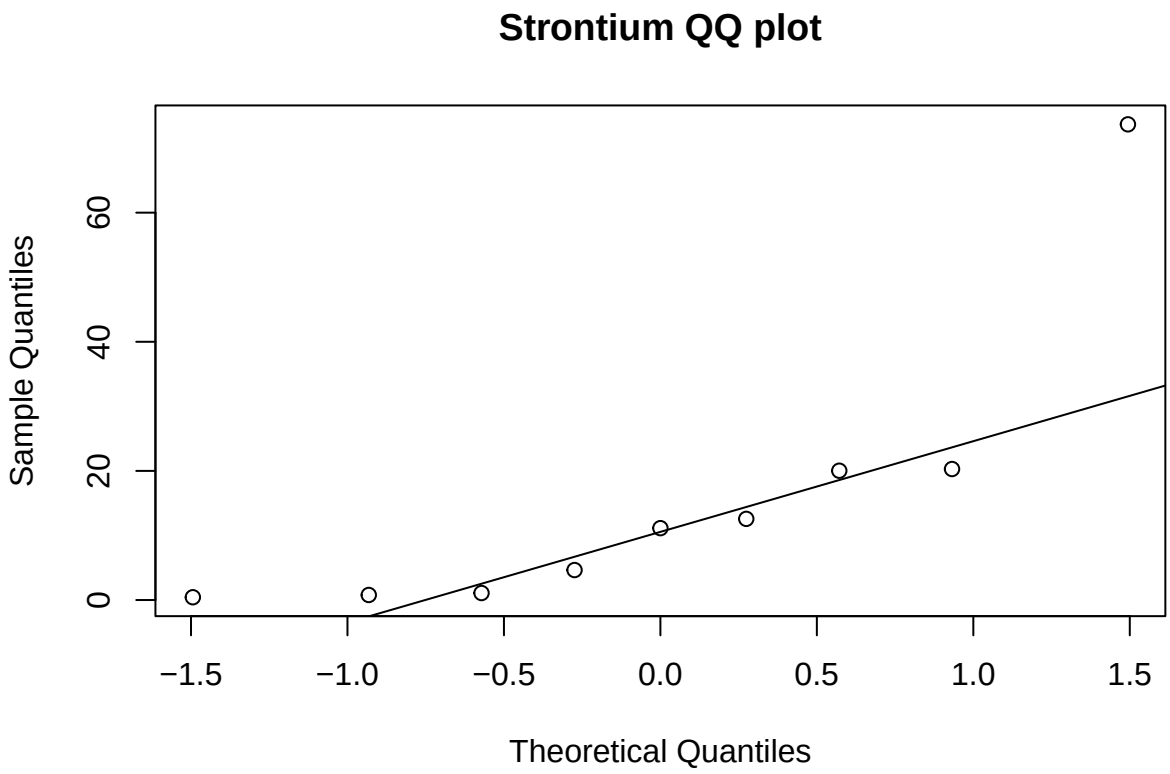
Given that T = 1.772481 $< \chi^2_{2,0.90}$, the point exists within the ellipse and is thus possible.

(c) Do these data appear to be bivariate normal? Discuss their status with reference to Q-Q plots and a scatter diagram. If the data are not bivariate normal, what implications does this have for the results in parts (a) and (b)?

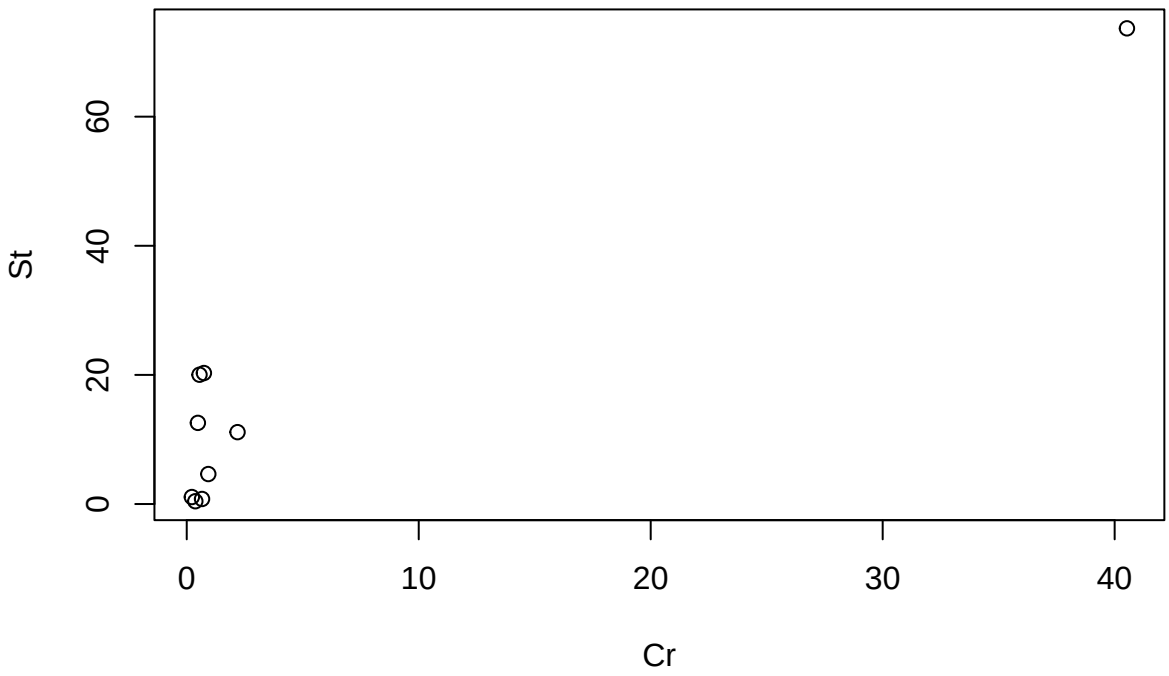
```
# Q-Q Plots
qqnorm(Cr, main="Chromium QQ plot")
qqline(Cr)
```




```
qqnorm(St, main="Strontium QQ plot")
qqline(St)
```




```
# Scatterplot
plot(
  Cr,
  St
)
```



The data does not appear to be bivariate normal with most observations packed to the left and an outlier very far to the right. This means that the results for (a) and (b) must be treated cautiously as they do not follow the necessary assumptions.

(d) Repeat the analysis with the obvious "outlying" observation removed. Do the inferences change? Comment.

```
hair2 <- hair[-2,]
hair2
```

```
##        Cr    St
## [1,] 0.48 12.57
## [2,] 2.19 11.13
## [3,] 0.55 20.03
## [4,] 0.74 20.29
## [5,] 0.66  0.78
## [6,] 0.93  4.64
## [7,] 0.37  0.43
## [8,] 0.22  1.08
```

```
cat("\n")
```

```
n <- 8
xbar2 <- colMeans(hair2)
xbar2
```

```
##      Cr      St
## 0.76750 8.86875
```

```
cat("\n")
```

```
s2 <- cov(hair2)
s2
```

```
##           Cr        St
## Cr 0.3785643  1.030282
## St 1.0302821 69.859755
```

```
cat("\n")
```

```
margin <- sqrt(
  qchisq(0.90, df = 2) * diag(s2) / n
)
cat("\n")
```

```
lower2 <- xbar2 - margin
upper2 <- xbar2 + margin
cat("\n")
```

```
cbind(lower2, upper2)
```

```
##       lower2    upper2
## Cr 0.3006819  1.234318
## St 2.5272590 15.210241
```

```
mu0 <- c(0.30,10)

d <- mu0 - xbar2

t <- n * t(d) %*% solve(s2) %*% d

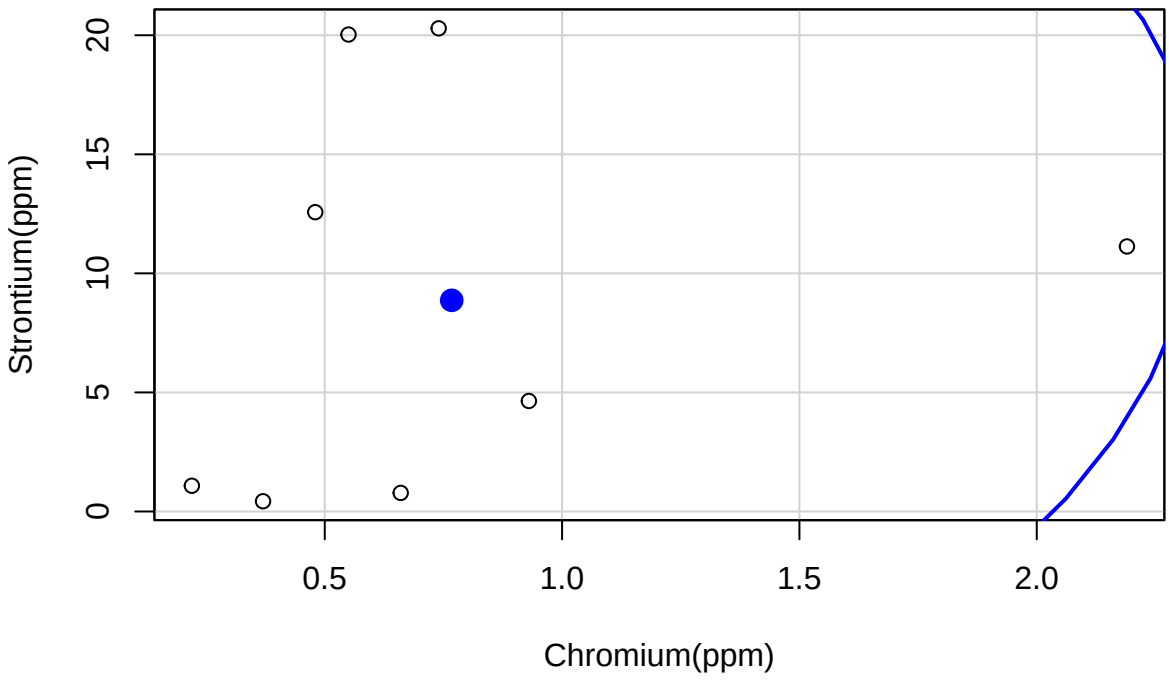
t
```

```
##          [,1]
## [1,] 5.307872
```

```
critical
```

```
## [1] 4.60517
```

```
dataEllipse(
  hair2[,1],
  hair2[,2],
  levels = 0.90,
  xlab = "Chromium(ppm)",
  ylab = "Strontium(ppm)"
)
```



Upon removing the outliers, we see changes in the results: The ellipse is much smaller than previous, [0.30, 10] is now outside the ellipse with a new t = 5.307872 < 4.60517, the critical value, 10 ppm strontium is also still possible within the 90% confidence interval.

4. Lumber Strength Measurements

```
x1 <- c(
  1232,1115,2205,1897,1932,1612,1598,1804,1752,2067,
  2365,1646,1579,1880,1773,
  1712,1932,1820,1900,2426,1558,1470,1858,1587,2208,
  1487,2206,2332,2540,2322
)

x2 <- c(
  4175,6652,7612,10914,10850,7627,6954,8365,9469,6410,
  10327,7320,8196,9709,10370,
  7749,6818,9307,6457,10102,7414,7556,7833,8309,9559,
  6255,10723,5430,12090,10072
)

lumber <- cbind(x1, x2)
```

(a) Construct and sketch a 95% confidence ellipse for the pair $[\mu_1, \mu_2]$', where $\mu_1 = \mathrm{E}(X_1)$ and $\mu_2 = \mathrm{E}(X_2)$.

```r
n <- nrow(lumber)

xbar <- colMeans(lumber)

S <- cov(lumber)

critical <- qchisq(0.95, df = 2)

theta <- seq(0, 2*pi, length.out = 500)

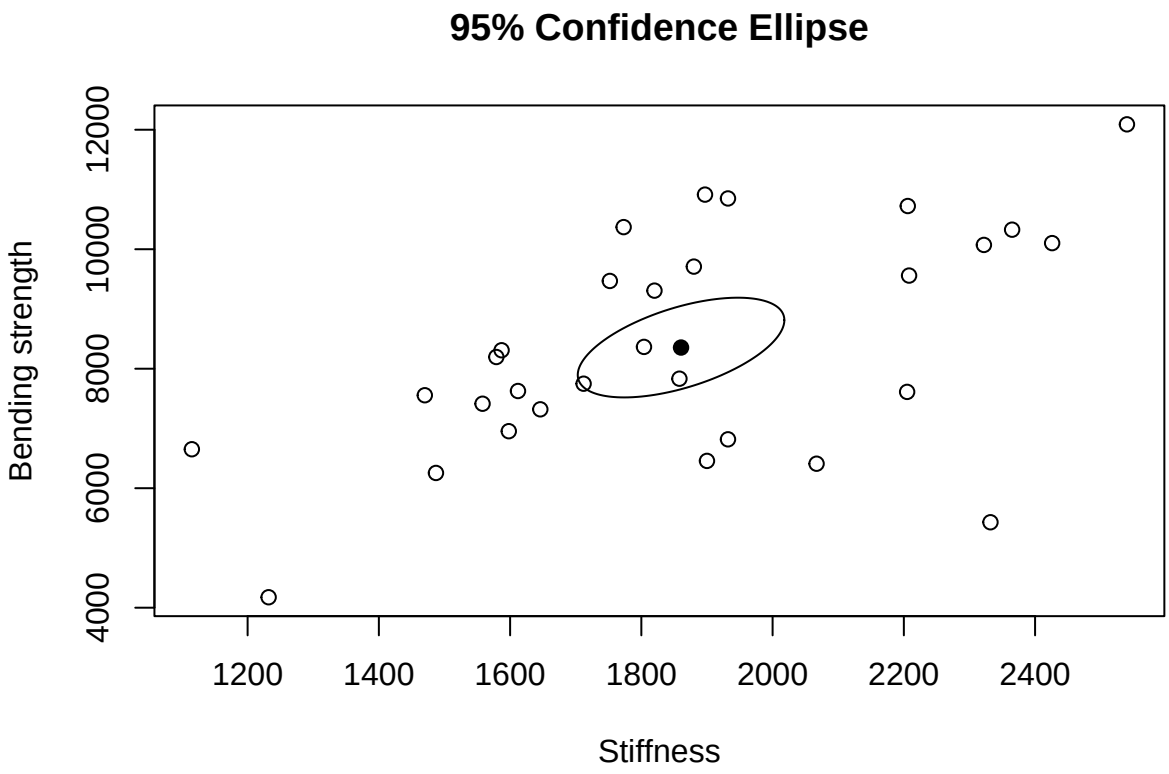
circle <- rbind(cos(theta), sin(theta))

ellipse <- xbar +
  sqrt(critical / n) *
  t(chol(S)) %*% circle

plot(
  x1, x2,
  xlab = "Stiffness",
  ylab = "Bending strength",
  main = "95% Confidence Ellipse"
)

lines(
  ellipse[1,],
  ellipse[2,]
)

points(
  xbar[1],
  xbar[2],
  pch = 19
)
```



(b) Suppose $\mu_1 0 = 2{,}000$ and $\mu_2 0 = 10{,}000$ represent "typical" values for stiffness and bending strength, respectively. Given the result in part (a), are the data consistent with these values? Explain.

```
mu0 <- c(2000, 10000)

d <- mu0 - xbar

T <- n * t(d) %*% solve(S) %*% d

T
```

```
##          [,1]
## [1,] 23.64778
```

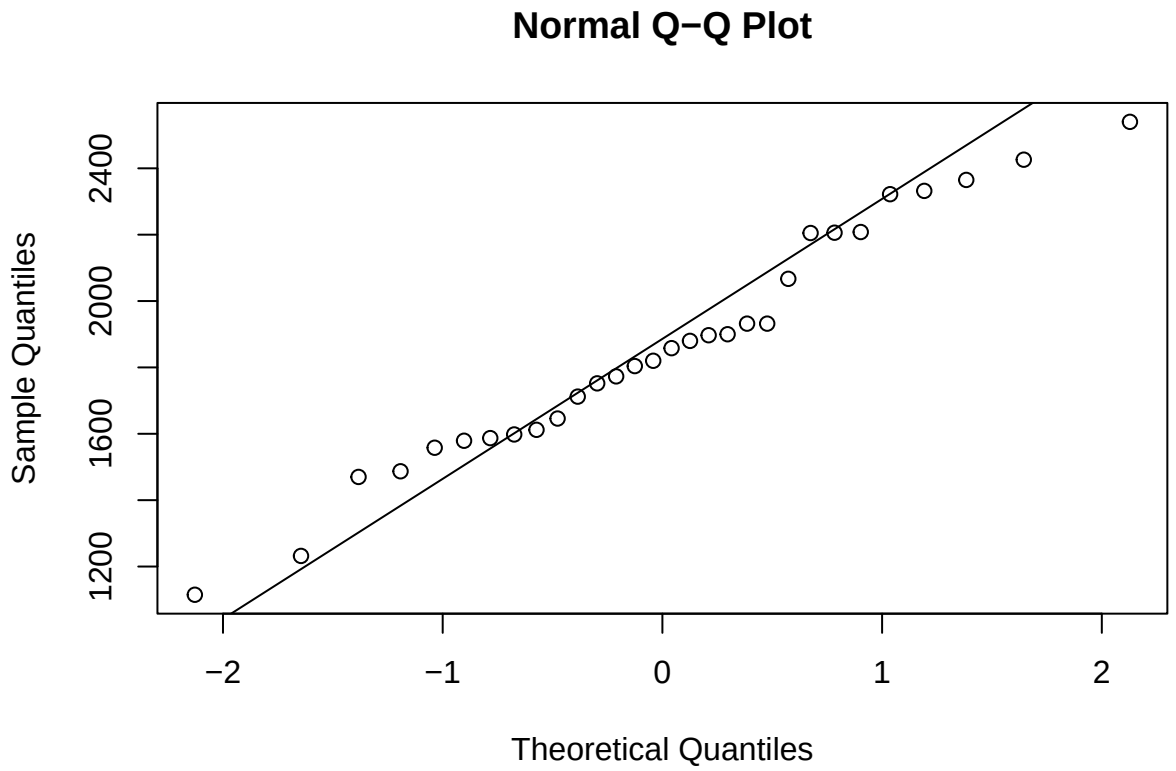
```
critical
```

```
## [1] 5.991465
```

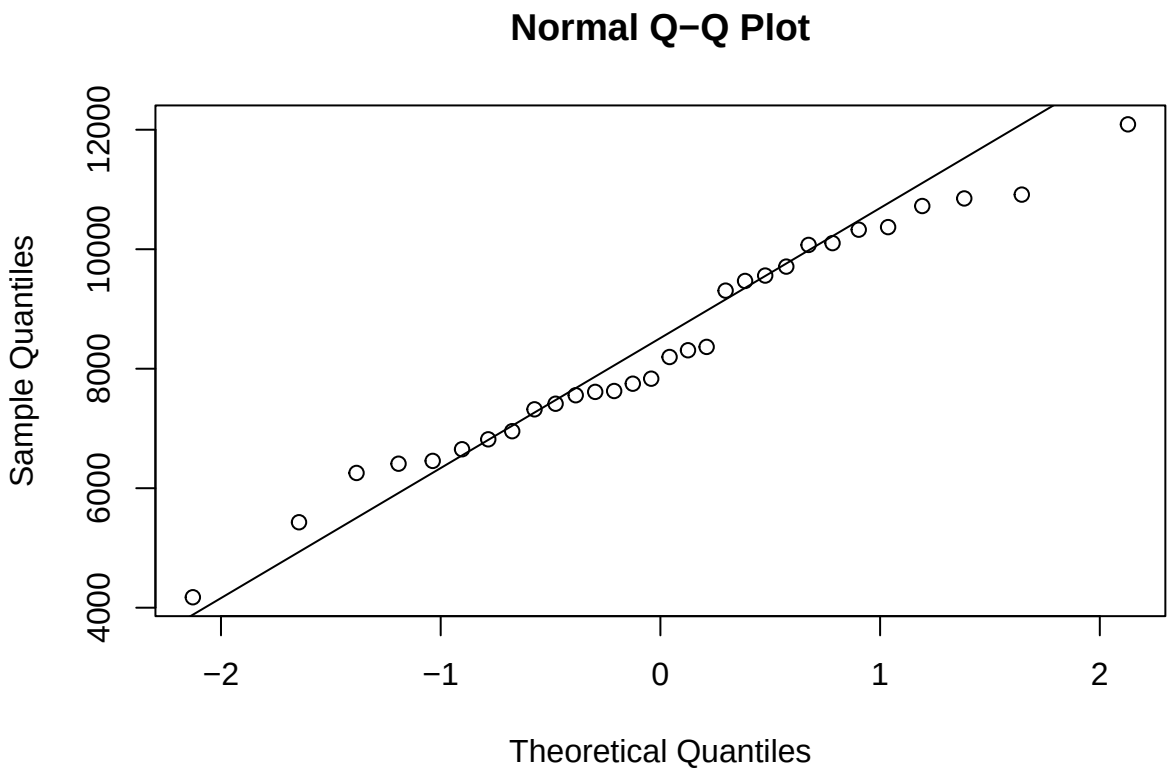
Since T = 23.64778 $> \chi^2_{2,0.95}$, then the proposed mean vector (2000, 10000)' is excluded from the 95% confidence ellipse region and is not consistent with the given data.

(c) Is the bivariate normal distribution a viable population model? Explain with reference to Q-Q plots and a scatter diagram.

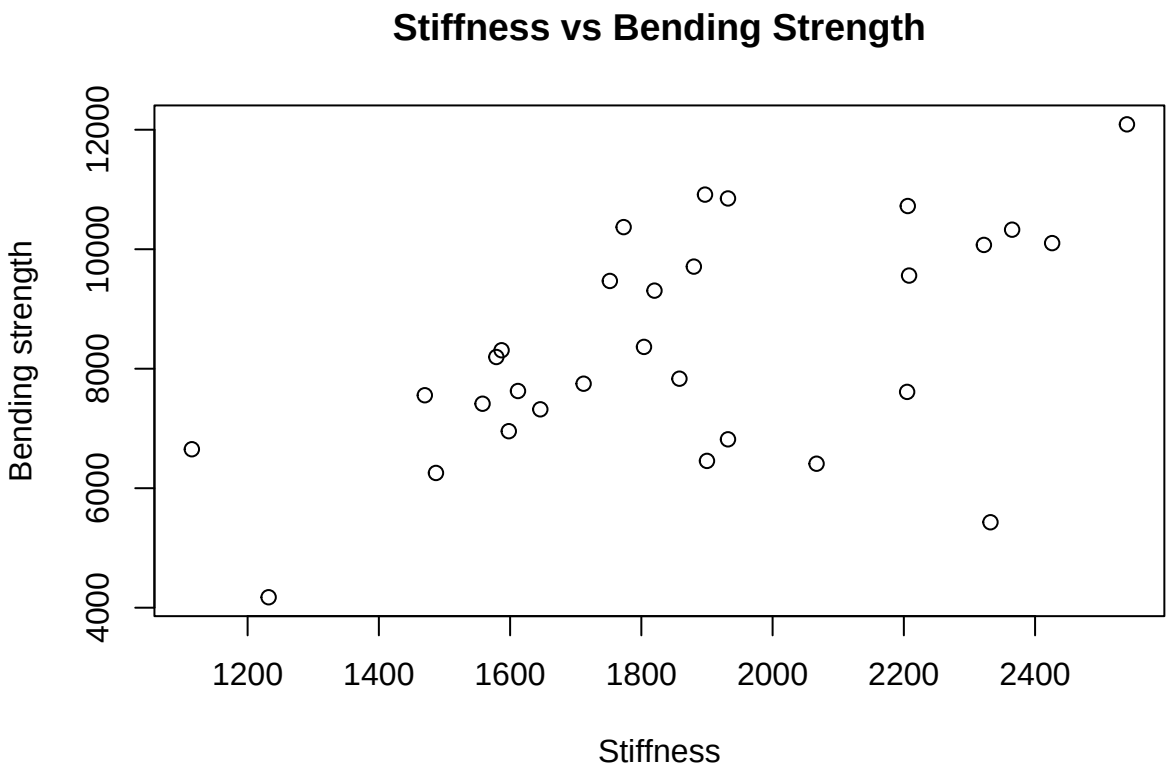
```
qqnorm(x1)
qqline(x1)
```




```
qqnorm(x2)
qqline(x2)
```




```r
# Scatterplot
plot(
  x1, x2,
  xlab = "Stiffness",
  ylab = "Bending strength",
  main = "Stiffness vs Bending Strength"
)
```



Both Q-Q plots showcase a straight diagonal line consistent with a normal distribution. Furthermore, the scatterplot does appear to have an elliptical shape. However, a sample size of n=30 may not be enough to be entirely conclusive.